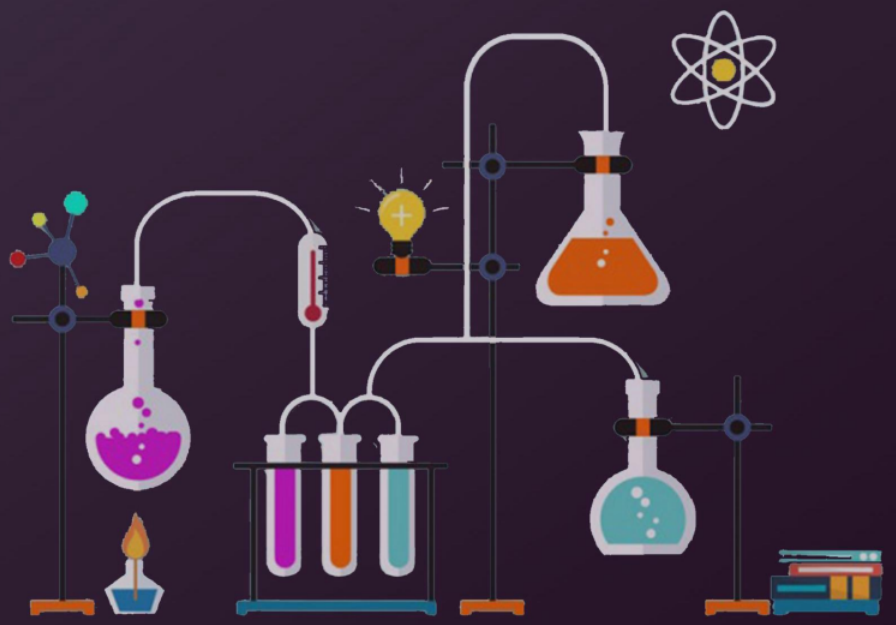
CHEMISTRY

Liquid Solution



SHEET

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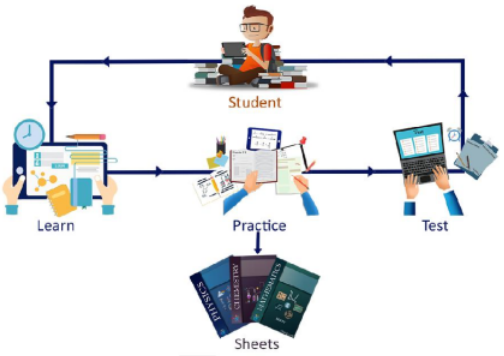
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Liquid Solution	1-3			
2	Solubility of Solid in Liquids	4,5	10,15,16	2	25
3	Henry's Law	6-9	17		
4	Raoult's Law	10-14	19	1-10	12,7
5	Ideal and Non-ideal Solutions	15-19	1,11	11	3,4,8
6	Liquid Vapour Equilibrium and Azeotropes	20-23	2,4-6,20	12,13	5,17
7	Colligative Properties-Relative Lowering of Vapour Pressure	24-28	21	14,15	26
8	Colligative Properties-Boiling Point Elevation	29-33	22-23	16-17,19	
9	Colligative Properties-Depression of Freezing Point-Part 1	34-36	24	31	
10	Colligative Properties-Depression of Freezing Point-Part 2	37-39	3	18,20-22	9-11
11	Osmosis	40-43		23-24	
12	Osmotic Pressure	44	25,26	25-30	
13	Abnormal Colligative Properties and Van't Hoff Factor	45-51	7	32-40	18-20, 27
14	ACC	52,53	8,9, 12-14,18	41-43	6,12-16, 21-24

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Liquid Solution

Q1. Substances that have unique, identical properties are called
(A) Homogeneous mixtures
(B) Pure Substance
(C) Mixtures
(D)Heterogeneoes mixtures

Q2. Air is Mixture of :
(A) Gases (B) Solutions
(C) Compound (D) Gases + liquid

Q3. A Solution is defined as a mixture that is ?
(A) Homologous
(B) Heterogeneous
(C) Homogeneous
(D) None of these

Solubility of Solid in Liquids

Q4. 0.1gram mole of urea dissolved in 100g of water.The molality of the solution is
(A) 1m (B) 0.01M
(C) 0.01m (D) 1.0M

Q5. Three statements are given about mole fraction
(i) Mole fraction of a solute + mole fraction of solvent = 1
(ii) Equal weights of Helium and methane are present in a gaseous mixture. The mole fraction of He is 4 / 5
(iii) The mole fraction of water in the aqueous solution of NaOH is 0.8. The molality of NaOH in the solution is nearly 14 moles kg^{-1}
(A) i and ii are correct
(B) ii and iii are correct
(C) i and iii are correct
(D) all are correct

Henry's Law

Q6. Henry's law constant (K_H) for methane in benzene at 298 K is 4.5×10^5 mm Hg. If we use the mole fraction of methane in the solution as a measure of its solubility then calculate the solubility of methane in benzene at 298 K under 760 mm Hg.
(A) 1.68×10^{-3} (B) 2.72×10^{-4}
(C) 3.24×10^{-5} (D) 1.68×10^{-2}

Q7. According to Henry law, the solubility of a gas in a given volume of liquid increase with increase in :-
(A) Temperature (B) Pressure
(C) Both (D) None

Q8. How many grams of CO_2 gas is dissolved in a 1 lt bottle of carbonated water if the manufacturer uses a pressure of 2.4 atmosphere in the bottling process at 25°C Given K_H of CO_2 water = $29.76 atm / mole / l$ at 25°C.
(A) 3.52 (B) 4.2 (C) 3.1 (D) 2.5

Q9. Incorrect statement is
(K_H = Henry's constant)
(A) K_H is a universal constant like Planck's constant.
(B) Higher is the value of K_H,lower is solubility of gas for a given partial pressure of gas
(C) K_H has temparature dependence
(D) K_H decreases with temparature

Raoult's Law

Q10. Mole fraction of A vapours above solution in mixture of A and B ($X_A = 0.4$) will be ? (P_A° = 100mm, P_B° = 200mm)
(A) 0.4 (B) 0.8
(C) 0.25 (D) 0.6

Q11. If Raoult's law is obeyed, the vapour pressure of the solvent in a solution is directly proportional to :

(A) Mole fraction of the solvent

(B) Mole fraction of the solute

(C) Mole fraction of the solvent and solute

(D) The volume of the solution

Q12. At a certain temperature pure liquid A and liquid B have vapour pressure 10 and 37 torr respectively. For a certain ideal solution of A and B, the vapours in equilibrium with the liquid has the components A and B in the partial pressure ratio $P_A : P_B = 1 : 7$. The mole fraction of A in the solution are ?

(A) 0.346 (B) 0.654

(C) 0.5 (D) 0.8

Q13. Which one of the following is the incorrect form of Raoult's law

(A) $\frac{P_s}{P^0} = \frac{N}{n+N}$ (B) $\frac{P^0}{P^0 - P_s} = 1 + \frac{N}{n}$

(C) $\frac{P^0 - P_s}{P_s} = \frac{n}{n+N}$ (D) $\frac{P_s}{P^o - P_s} = \frac{N}{n}$

Q14. At 300 K, the vapour pressure of an ideal solution containing 3 mole of A and 2 mole of B is 600 torr. At the same temperature, if 1.5 mole of A & 0.5 mole of C (non-volatile) are added to this solution the vapour pressure of solution increases by 30 torr. What is the value of P_B^o ?

(A) 940 (B) 405

(C) 90 (D) none of these

Ideal and Non-ideal solutions

Q15. Which condition is not satisfied by an ideal solution

(A) $\Delta H_{mixing} = 0$

(B) $\Delta V_{mixing} = 0$

(C) $\Delta S_{mixing} = 0$

(D) Obeyance of Raoult's law

Q16. A solution containing components A and B follows Raoult's law :-

(A) A – B attraction force is greater than A – A and B – B

(B) A – B attraction force is less than A – A and B – B

(C) A – B attraction force remains same as A – A and B – B

(D) Volume of solution is different from sum of volume of solute and solvent

Q17. Which of the following is incorrect for an ideal solution :-

(A) $\Delta H_{mix} = 0$ and $\Delta V_{mix} = 0$

(B) $\Delta V_{mix} = 0$ and $\Delta S_{mix} > 0$

(C) $\Delta H_{mix} > 0$ and $\Delta S_{mix} > 0$

(D) $\Delta G_{mix} < 0$ and $\Delta S_{mix} > 0$

Q18. Which of the following does not show negative deviation from Rault's law ?

(A) Acetone – Chloroform

(B) Toluene – Benzene

(C) Nitric Acid – Water

(D) Chloroform – Benzene

Q19. Identify the false statement :

(A) The volume of a solution cannot be less than the sum of volumes of the pure solvent and solute used to prepare the solution (binary solution of two liquids).

(B) At constant T and P, ΔG_{mix} will be necessarily negative for an ideal solution.

(C) An ideal binary solution $(p_A^0 \neq p_B^0)$ cannot form an azeotropic mixture.

(D) In binary solutions ideality is more of an exception rather than a rule.

Liquid Vapour Equilibrium and Azeotropes

Q20. Azeotropic mixture are ?

(A) Mixture of two solids

(B) Those which boil at different temperatures

(C) Those which can be fractionally distilled

(D) Constant boiling mixtures

Q21. An azeotropic mixture of two liquids boil at a lower temperature than either of them when -

(A) It is saturated

(B) It does not deviate from Raoult's law

(C) It shows negative deviation from Raoult's law

(D) It shows positive deviation from Raoult's law

Q22. The vapour pressure of water depends upon.

(A) Surface area of container

(B) Volume of container

(C) Temperature

(D) All

Q23. X-is a non-volatile solute and Y is a volatile solvent.The following vapoure pressure are observed by dissolving X in Y.

X/mol lit^{-1}	Y/mm of Hg
0.10	P_1
0.25	P_2
0.01	P_3

The correct order of vapoure pressures is -

(A) $P_1 < P_2 < P_3$ (B) $P_3 < P_2 < P_1$

(C) $P_3 < P_1 < P_2$ (D) $P_2 < P_1 < P_3$

Colligative Properties-Relative Lowering of Vapour Pressure

Q24. The solubility of a specific non-volatile salt is 4 g in 100 g of water at 25°C. If 2.0 g, 4.0 g and 6.0 g of the salt added of 100 g of water at 25°C, in system X, Y and Z. The vapour pressure would be in the order :-

(A) X < Y < Z (B) X > Y > Z

(C) Z > X = Y (D) X > Y = Z

Q25. Colligative properties of the solution depend upon

(A) Nature of the solution

(B) Nature of the solvent

(C) Number of solute particles

(D) Number of moles of solvent

Q26. Which is not a colligative property ?

(A) Osmotic pressure

(B) Lowering in vapour pressure

(C) Depression in freezing point

(D) Refrective index

Q27. When 25 grams of a non-volatile solute is disolved in 100 grams of water, the vapour pressure is lowered by 2.25×10^{-1} mm.If the vapour pressure of water at 20°C is 17.5mm,What is the molecular weight of the solute?

(A) 206 (B) 302

(C) 350 (D) 276

Q28. A solution is obtained by dissolving 0.2 moles of urea in a litre water. Another solution is obtained by dissolving 0.4 moles of cane-sugar in a litre of water at the same temperature. The lowering of vapour pressure to the first solution is -

(A) Same as that of the second solution

(B) Half to that of the second solution

(C) Double to that of the second solution

(D) None

Colligative Properties- Boiling Point Elevation

Q29. The molal elevation constant is the ratio of the elevation in B.P. to

(A) Molarity

(B) Molality

(C) Mole fraction of solute

(D) Mole fraction of solvent

Q30. Arrange the following aqueous solutions in the order of their increasing boiling points :-

(i) 10^{-4} M NaCl (ii) 10^{-4} M Urea

(iii) 10^{-3} M $MgCl_2$ (iv) 10^{-2} M NaCl

(A) (i) < (ii) < (iv) < (iii)

(B) (ii) < (i) = (iii) < (iv)

(C) (ii) < (i) < (iii) < (iv)

(D) (iv) < (iii) < (i) = (ii)

Q31. The boiling point of C_6H_6, CH_3OH, $C_6H_5NH_2$ and $C_6H_5NO_2$ are 80^0C, 65^0C, 184^0C and 212^0C respectively which will show highest vapour pressure at room temperature :

(A) C_6H_6 (B) CH_3OH

(C) $C_6H_5NH_2$ (D) $C_6H_5NO_2$

Q32. The rise in the boiling point of a solution containing 1.8g of glucose in 100g of solvent is 0.1^0C. The molal elevation constant of the liquid is -

(A) 0.01 K/m (B) 0.1 K/m

(C) 1K/m (D) 10 K/m

Q33. If the elevation in boiling point of a solution of 10gm of solute (mol. wt. = 100) in 100 gm of water is ΔT_b, the ebullioscopic constant of water is -

(A) 10 (B) $10\Delta T_b$

(C) ΔT_b (D) $\frac{\Delta T_b}{10}$

Colligative Properties -Depression of Freezing Point -Part-1

Q34. If K_f value of H_2O is 1.86. The value of ΔT_f for 0.1 m solution of non-volatile solute is -

(A) 18.6 (B) 0.186

(C) 1.86 (D) 0.0186

Q35. A 5% solution(by mass) of cane sugar in water has freezing point of 271K.Calculate the freezing point of 5% glucose in water if freezing point of water is 273.15K.

(A) 269.07 k (B) 269.07°C

(C) 373k (D) 373°C

Q36. 1.00g of a non-electrolyte solute dissolved in 50g of benzene lowered the freezing point of benzene by 0.40K.The freezing point depression constant of benzene is 5.12K kg mol^{-1}.The molar mass of the solute is -

(A) 128 g mol^{-1} (B) 256 g mol^{-1}

(C) 64 g mol^{-1} (D) 32 g mol^{-1}

Colligative Properties- Depression Of Freezing Point- Part2

Q37. The normal freezing point of nitro benzene is 278.82K. A 0.25 molal solution of solute A in nitrobenzene causes the freezing point to decrease further to 276.82K. The value of K_f for nitrobenzene is :-

(A) 2 K $kgmol^{-1}$ (B) 4 K $kgmol^{-1}$

(C) 6 K $kgmol^{-1}$ (D) 8 K $kgmol^{-1}$

Q38. The depression in freezing point of 0.1 molal aqueous solution of HCl, $CuSO_4$ and H_2SO_4 are in the ratio :

(A) 1 : 1 : 1 (B) 1 : 2 : 3

(C) 1 : 1 : 1.5 (D) 2 : 4 : 3

Q39. How many grams of methyl alcohol should be added to 10 litre tank of water to prevent it from freezing at 268 K ?

(K_f for water is 1.86 K Kg mol^{-1})

(A) 880.07g (B) 899.04g

(C) 886.02g (D) 868.06 g

Osmosis

Q40. Pure water can be obtain from sea water by :

(A) Centrifugation (B) Plasmolysis

(C) Reverse osmosis (D) Sedimentation

Q41. In osmosis phenomenon :

(A) Solvent molecules move from higher concentration to lower concentration

(B) Solvent molecules move from lower concentration to higher concentration

(C) Solute molecules move from higher concentration to lower concentration

(D) Solute molecules move from lower concentration to higher concentration

Q42. The plant cell will shrink when placed in :

(A) Water

(B) A hypotonic solution

(C) A hypertonic solution

(D) An isotonic solution

Q43. The best colligative property used for the determination of molecular masses of polymers is ?

(A) Relative lowering in vapour pressure
(B) Osmotic pressure
(C) Elevation in boiling point
(D) Depression in freezing point

Osmotic Pressure

Q44. A solution of glucose $(C_6H_{12}O_6)$ is isotonic with 4g of urea per litre of solution. The concentration of glucose is -

(A) 4 g/L (B) 8 g/L
(C) 12 g/L (D) 14 g/L

Abnormal Colligative Properties and van't Hoff Factor

Q45. Mark the correct relationship between the boiling points of very dilute solutions of $BaCl_2$ (T_1) and KCl (T_2), having the same molarity.

(A) $T_1 = T_2$
(B) $T_1 > T_2$
(C) $T_2 > T_1$
(D) T_2 is approximately equal to T_1

Q46. Which solution will have least vapour pressure ?

(A) 0.1 M $BaCl_2$ (B) 0.1 M urea
(C) 0.1 M Na_2SO_4 (D) 0.1 M Na_3PO_4

Q47. The substance A when dissolved in solvent B shows the molecular mass corresponding to A_3. The Vant Hoff's factor will be ?

(A) 1 (B) 2
(C) 3 (4) 1/3

Q48. When common salt is dissolved in water

(A) Melting point of the solution increases
(B) Boiling point of the solution increases
(C) Boiling point of the solution decreases
(D) Both Melting point and Boiling point is decreases

Q49. 7.6 gm KBr in 1250 ml solution was found to have an osmotic pressure of 1.804 atm at 27°C. Calculate degree of dissociation in % (At.wt. K = 39, Br = 80)

(A) 43.4% (B) 46.4%
(C) 4.64% (D) None

Q50. Consider the two solutions :-

(I) 0.5 M NaCl aqueous solution at 25°C ; NaCl is completely ionized

(II) 2.0 M C_6H_5COOH in benzene at 25°C, the acid dimerizes to the full extent.

Which of the following statement is/are correct :-

(A) Both the solutions display equal osmotic pressures
(B) Both have equal vapour pressures
(C) Solution II is hypertonic
(D) Solution II is hypotonic

Q51. If in solvent, n simple molecules of solute combine to form an associated molecule, X is degree of association the Van't Hoff's factor 'i' is equal to :

(A) $\frac{1}{1-nx}$ (B) $\frac{1-x+nx}{1}$

(C) $\frac{1-x+\frac{x}{n}}{1}$ (D) $\frac{\frac{x}{n}-1+x}{1}$

ACC

Q52. On supplying pressure in water $\rightleftharpoons$ vapour physical equilibrium :-

(A) Boiling point increases
(B) Melting point decreases
(C) Boiling point decreases
(D) Melting point increases

Q53. When 20g of naphthoic acid $(C_{11}H_8O_2)$ is dissolved in 50g of benzene (K_f = 1.72 $kgmol^{-1}$), a freezing point depression of 2K is observed. The Van't Hoff factor (i) is ?

(A) 0.5 (B) 1
(C) 2 (D) 3

Exercise - 1A
Concept Builder

ONE OR MORE THAN ONE CORRECT TYPE

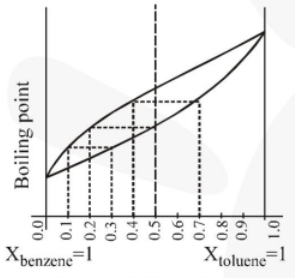
*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following is correct for a non-ideal solution of liquids A and B, showing negative deviation?

(A) ΔH_{mix} = –ve (B) ΔV_{mix} = –ve

(C) ΔS_{mix} = +ve (D) ΔG_{mix} = –ve

Q2. The following graph represents variation of boiling point with composition of liquid and vapours of binary liquid mixture. The graph is plotted at constant pressure, for a solution of mole fraction 0.5



Which of the following statement(s) is **correct**.

(A) $X_{benzene}$ = 0.5 and $Y_{toluene}$ = 0.2

(B) $X_{toluene}$ = 0.3 and $Y_{benzene}$ = 0.6

(C) $X_{benzene}$ = 0.3 and $Y_{toluene}$ = 0.4

(D) $X_{benzene}$ = 0.7 and $Y_{toluene}$ = 0.3

Q3. In the depression of freezing point experiment, it is found that the –

(A) vapour pressure of the solution is less than that of pure solvent

(B) vapour pressure of the solution is more than that of pure solvent

(C) only solute molecules solidify at the freezing point

(D) only solvent molecules solidify at the freezing point

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

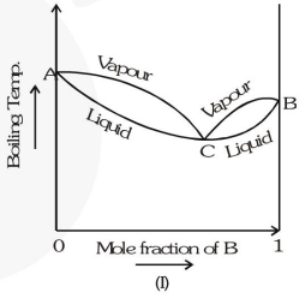
Paragraph for (Q. 4 to 6)

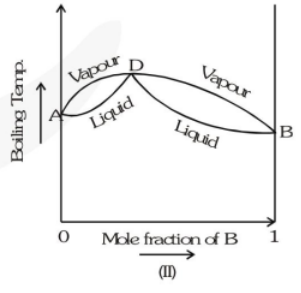
Azoetropes are constant boiling mixtures, which like a pure chemical compound boils at a constant temperature and distills over completely at same temperature without change in composition. There may be two type of binary solution which gives azeotropic mixture.

Type-I : Mixture showing minimum in "Boiling point-Composition curve".

Type-II: Mixture which shows a maxima in "Boiling point-Composition curve".

Both types are shown below :





Q4. For figure (I) and (II) predict which of the component (i.e., A and B) is more volatile ?

(A) Liquid A

(B) Liquid B

(C) Both have equal volatility

(D) Can't predict

Q5. Which of the type above (i.e., Type-I or Type-II) shows negative deviation from Raoult's law ?

(A) Solution of Type-I

(B) Solution of Type-II

(C) Both Type-I and Type-II

(D) None of the above

Q6. Let the solution of "water-ethanol" corresponds to Type-I and at point 'C', it had a composition of 95.6% of by weight in vapour phase". If this solution is boiled at 78.13°C then find out the % by wt. of water in 'liquid phase' (Given T_C (temperature at point C) = 78.13°C :-

(A) 95.6% (B) 4.4%

(C) 72.4% (D) 63%

Paragraph for (Q. 7 to 9)

A dilute solution contains 'x' moles of solute A in 1 kg of solvent with molal elevation constant K_b. The solute dimerises in the solution according to the following equation. The degree of association is (α) is $2A \rightleftharpoons A_2$.

Q7. The van't Hoff factor will be :

(A) $i = 1 - 2\alpha$ (B) $i = 1 - \frac{\alpha}{2}$

(C) $i = 1 + \frac{\alpha}{2}$ (D) $i = 1 + \alpha$

Q8. The molecular weight observed will be :

(A) greater than actual molecular weight

(B) lesser than actual molecular weight

(C) equal to the actual molecular weight

(D) cannot be predicted by the data given

Q9. The degree of association is equal to :

(A) $\alpha = \frac{(K_b x - \Delta T_b)}{\Delta T_b . 2}$

(B) $\alpha = \frac{2(K_b x - \Delta T_b)}{K_b . x}$

(C) $\alpha = 2 + \frac{2\Delta T_b}{K_b x}$

(D) $\alpha = \frac{\Delta T_b}{2K_b x}$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q10. Match the laws listed in column I with the corresponding equations given in column II.

Column - I	**Column - II**
(A) Normality	(p) Vary with temperature
(B) Molarity	(q) Independent of temperature variation
(C) Molality	(r) Involves ratio
(D) Mole fraction	(s) Involves weight

Q11. For a binary liquid solution of A and B. P_A^0 = pure vapour pressure of A. P_B^0 = pure vapour pressure of B. X_A = mole fraction of A in liquid phase. Y_A = mole fraction of A in vapour phase.

Column-I	**Column-II**
(A) $P_A^0 > P_B^0$ [Ideal liquid solution]	(p) $X_A = Y_A$
(B) Azeotropic mixture	(q) $X_A < Y_A$
(C) Equimolar ideal mixture of A & B with $P_A^0 < P_B^0$	(r) $X_A > Y_A$
(D) Equimolar ideal mixture of A & B with $P_A^0 = P_B^0$	(s) $Y_B > Y_A$
	(t) $Y_B = Y_A$

Q12. Match the following :

Column - I	Column - II
(A) A nearly ideal solution	(p) $H_2O + C_2H_5OH$
(B) A solution showing positive deviation	(q) $C_6H_6 + C_6H_5CH_3$
(C) A solution showing negative deviation	(r) $C_6H_6 + CHCl_3$
(D) An azeotropic mixture with high.B.P	(s) Acetone + C_2H_5OH
	(t) ΔH & ΔV are +ve
	(u) ΔH & ΔV are -ve

Q13. Match the column :

Column I	Column II
(A) Elevation of B.P.	(p) Colligative property
(B) Osmotic pressure	(q) Ebullioscopic constant
(C) Relative lowering in V.P.	(r) Berkeley-Heartley
(D) Depression of F.P.	(s) Ostwald and walker method

Q14.

Column I	Column II
(A) ppm	(p) Van't Hoff factor
(B) ΔT_b	(q) Molal depression constant
(C) K_f	(r) Elevation in boiling point
(D) i	(s) $\frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6$

SUBJECTIVE TYPE

Q15. If 0.4 g of NaOH is present in 40 mL of solution. What is the molarity and normality. [Molecular mass of NaOH = 40]

Q16. The normality of 1.5M H_3PO_4 is -

Q17. Henry's Law Constant for CO_2 in water is 1.67×10^8 Pa at 298K. Calculate the quantity of CO_2 in 1 L of soda water when packed under 2.5 atm CO_2 pressure at 298 K.

Q18. Methanol and C_2H_5OH form an ideal solution. Solution is prepared by mixing 32 gm of CH_3OH and 23 gm of C_2H_5OH at 300K. At 300K $P^0_{methanol} = 90$ mm of Hg and $P^0_{ethanol} = 51$ mm of Hg.
Calculate :
(i) Partial vapour pressure of its constitutents and total vapour pressure of solution.
(ii) Mole fraction of each component in vapour phase.

Q19. The vapour pressure of pure liquid A is 40 torr at 310 K. The vapour pressure of this liquid in solution with liquid B is 32 torr. Calculate X_A in solution.

Q20. At 88°C benzene has a vapour pressure of 900 torr and toluene has a vapour pressure of 360 torr. What is the mole fraction of benzene in the mixture with toluene that will be boil at 88°C at 1 atm pressure, benzene – toluene form an ideal solution.

Q21. 1 mole heptane (V.P. = 92 mm of Hg) is mixed with 4 mol. Octane (V.P. = 31 mm of Hg), form an ideal solution. Find out the vapour pressure of solution.

Q22. What will be the boiling point of bromine when 174.5 mg of octa-atomic sulphur is added to 78 g of bromine. K_b for Br_2 is 5.2 K mol^{-1} kg and boiling point of Br_2 is 332.15 K.

Q23. The rise in boiling point of a solution containing 1.8 g glucose in 100 g of a solvent is 0.1^0C. The molal elevation constant of the liquid is -

Q24. If freezing point of a solution prepared from 1.25 g of a non electrolyte and 20 g of water is 271.9 K, the molar mass of the solute will be :

Q25. A solution containing 8.6 g urea in one litre was found to be isotonic with 0.5% (wt./vol.) solution of an organic, non-volatile solution. The molecular weight of latter is -

Q26. At 27° C, 36 g of glucose per litre has an O. P. of 4.92 atm. If the osmotic pressure of solution is 1.5 atm at the same temperature, what should be its concentration ?

Exercise - 2 — *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Two liquid A and B have vapour pressure in the ratio $P_A^0 : P_B^0 = 1:3$ at a certain temperature. If the ratio of mole fractions of A to B in the vapour phase is 4 : 3, then the mole fraction of B in the solution at the same temperature is ?

(A) $\frac{1}{5}$ (B) $\frac{2}{3}$

(C) $\frac{4}{5}$ (D) $\frac{1}{4}$

Q2. The mole fraction of the solute in one molal aqueous solution is ?

(A) 0.027 (B) 0.036

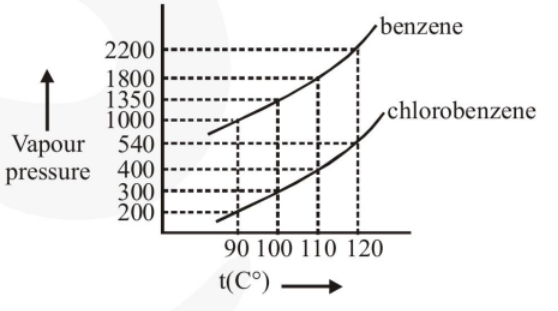
(C) 0.018 (D) 0.009

Q3. A solution has 1 : 4 mole ratio of pentane to hexane. The vapour pressures of the pure hydrocarbons at 20°C are 440 mm Hg for pentane and 120 mm Hg for hexane. The mole fraction of pentane in the vapour phase would be ?

(A) 0.200 (B) 0.478

(C) 0.549 (D) 0.789

Q4. Determine the boiling point of a mixture containing 1560 g benzene (molar mass = 78) and 1125 g chlorobenzene (molar mass = 112.5) using the following graph at atmospheric pressure of 1000 torr :-



(A) 90° (B) 100°C

(C) 110°C (D) 120°C

Q5. Calculate the pure vapour pressure of liquid A and liquid B,

Given : $P_s = 300 - 175\ X_B$

Where, X_B = made fraction of liquid B :-

(A) 125, 300

(B) 300, 125

(C) 25, 175

(D) None of these

Q6. At 25°C, the vapour pressure of pure liquid A (mol. wt. = 40) is 100 torr, while that of pure liquid B (mol. wt. = 80) is 40 torr. The vapour pressure at 25°C of a solution containing 20 gm of each A and B is :-

(A) 59.8 torr (B) 68 torr

(C) 80 torr (D) None

Q7. Two liquids A and B have P_A^o and P_B^o in the ratio of 1 : 3 and the ratio of number of moles of A and B in liquid phase are 1 : 3 then mole fraction of A' in vapour phase in equilibrium with the solution is equal to :-

(A) 0.1 (B) 0.2

(C) 0.5 (D) 1.0

Q8. Benzene and toluene form nearly ideal solutions. At 20°C, the vapour pressure of benzene is 75 torr and that of toluene is 22 torr. The Partial vapour pressure of benzene at 20°C for a solution containing 78g of benzene and 46g of toluene in torr is ?

(A) 50 (B) 25

(C) 37.5 (D) 53.5

Q9. At 25°C, the total pressure of a solution obtained by mixing 3 moles of A and 2 moles of B is 184 torr. What is the vapour pressure (in torr) of pure B at the same temperature ? (Vapour pressure of pure A, at 25°C is 200 torr) :-

(A) 180 (B) 160

(C) 16 (D) 100

Q10. A binary solution of liquid A and B will show positive deviation from Raoult's law if it fulfils the following condition :-

(i) $P_A > P_A^0 X_A$ and $P_B > P_B^0 X_B$

(ii) The inter molecular force of

A–B < A–A, B–B

(iii) ΔH_{mix} is positive

(iv) ΔV_{mix} is negative

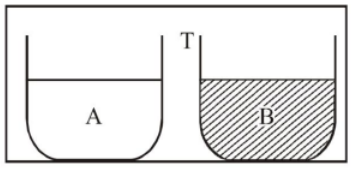
(A) I, II and IV are True

(B) I, II are wrong

(C) I, II, III are true

(D) I, III, IV are true

Q11. Two beakers A and B containing pure water and aqueous solution are shown. What will be observed gradually with passage of time ?



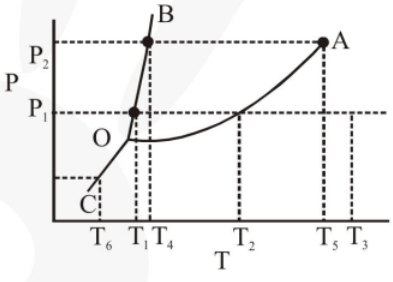
(A) Level of water in beaker A becomes lower than that in beaker B

(B) Level of water in beakers A and B remains constant

(C) Level of water in beaker A increases and that in beaker B decreases

(D) Level of both the beakers increases

Q12. Select the incorrect statement using phase diagram of a substance



(A) At pressure P_2, T_4 is melting point

(B) Substance will be in liquid state if temperature T is '$T_4 < T < T_5$' at P_2 pressure

(C) At T_3 temperature vapour pressur of liquid is P_1

(D) Sublimation temperature is always less than triple point

Q13. Which of the following plots does not represent the behaviour of an ideal binary liquid solution

(A) Plot of P_A versus X_A (mole fraction of A in liquid phase) is linear

(B) Plot of P_B versus X_B is linear

(C) Plot of p_{total} versus X_A (or X_B) is linear

(D) Plot of p_{total} versus X_A is non linear

Q14. The relative lowering of vapour pressure produced by dissolving 71.5 g of a substance in 1000 g of water is 0.00713. The molecular weight of the substance will be ?

(A) 18.0 (B) 342

(C) 60 (D) 180

Q15. One mole of non volatile solute is dissolved in two moles of water. The vapour pressure of the solution relative to that of water is ?

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) $\frac{3}{2}$

Q16. When 1 mole of a solute is dissolved in 1 kg of H_2O, boiling point of solution was found to be 100.5°C. K_b for H_2O is ?

(A) 0.5 (B) 100

(C) 100.5 (D) 95.5

Q17. The molal elevation constant of water is 0.51. The boiling point of 0.1 molal aqueous NaCl solution is nearly :

(A) 100.05 °C (B) 100.1 °C

(C) 100.2° C (D) 101.0° C

Q18. K_f for water is 1.86 K kg mol^{-1}. If automobile radiator holds 1.0 kg of water, how many grams of ethylene glycol ($C_2H_6O_2$) must be added to get the freezing point of the solution lowered to –2.8°C ?

(A) 27g (B) 72 g

(C) 93 g (D) 39 g

Q19. 0.15 gm of a substance dissolved in 15 gm of solvent, boiled at a temp. higher by 0.216°C than that of the pure solvent. Find out the molecular weight of the substance. (Molal elevation constant is 2.16°C) :-

(A) 1.01 (B) 10.0

(C) 100 (D) 10

Q20. Calculate the molecular weight of a substance which forms a 7.0% by mass solution in water which freezes at – 0.93°C. The cryoscopic constant of water is 1.86°C kg mol^{-1}:-

(A) 140 g mol^{-1} (B) 150.5 g mol^{-1}

(C) 160 g mol^{-1} (D) 155 g mol^{-1}

Q21. Arrange the following aqueous solutions in order of increasing freezing points (that is lowest first):-

I. 0.10 m $Ba_3(PO_4)_2$

II. 0.10 m Na_2SO_4

III. 0.10 m C_2H_5OH

IV. 0.10 m KCl

V. 0.10 m Li_3PO_4

(A) I < V < II < IV < III

(B) III ≡ IV < II < V < I

(C) IV < II < V < I < III

(D) I < V < II < IV = III

Q22. The decrease in the freezing point of an aqueous solution of a substance is 1.395 K and that in the freezing point of benzene solution of the same substance is 1.280 K. Explain the difference in ΔT. The substance :

(A) dissociates in the aqueous solution as well as in the benzene solution

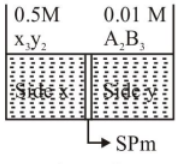
(B) forms complexes in solution

(C) associates in the benzene solution

(D) dissociates in the aqueous solution and not in the benzene solution

Q23. X_3Y_2 (i = 5) when reacted with A_2B_3 (i = 5) in aqueous solution given brown colour.

These are seperated by a semipermeable membrane AB as shown. Due to osmosis there is :-



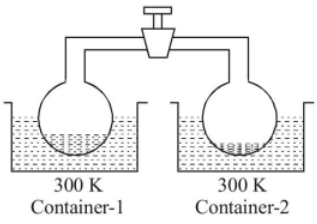
(A) Brown colour formation in side x

(B) Brown colour formation in side y

(C) Formation in both of sides x and y

(D) No brown colour formation

Q24. Two container are connected as shown in diagram.



Container-1 contains 0.1 M urea solution and container-2 contains 0.1 M $Al_2(SO_4)_3$ solution. If stopper is opened for sufficient time and then closed then :

(A) Weight of container-2 containing $Al_2(SO_4)_3$ increases.

(B) Weight of container-2 containing $Al_2(SO_4)_3$ decreases.

(C) No change in weight occurs in two container.

(D) Weight change if occurs can not be predicted.

Q25. The osmotic pressure of a solution at 0°C is 2 atm. What will be its osmotic pressure at 273°C under similar conditions ?

(A) 0.5 atm. (B) 2×273 atm.

(C) 4 atm. (D) 273/2 atm.

Q26. Insulin is dissolved in a suitable solvent and the osmotic pressure (π) of solutions of various concentration (g/ cm^3) C is measured at 20°C. The slope of a plote of π against C is found to be 4.65×10^{-3}. The molecular weight of the insulin is -

(A) 3×10^5 (B) 9×10^5

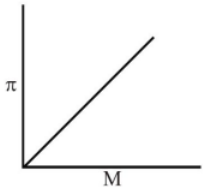
(C) 4.5×10^5 (D) 5.16×10^6

Q27. pH of a 0.1 M monobasic acid is measured as 2. Its osmotic pressure at a given temperature T K is -

(A) 0.1 RT (B) 0.11 RT

(C) 1.1 RT (D) 0.01 RT

Q28. A graph of osmotic pressure (π) vs molarity (M) of aqueous solution of NaCl at room temperature is given below. The slope of the line represents :-



(A) Concentration

(B) Universal gas constant

(C) RT

(D) 2RT

Q29. Osmotic pressure of 30% solution of glucose is 1.20 atm and that of 3.42% solution of cane sugar is 2.5 atm. The osmotic pressure of the mixture containing equal volumes of the two solutions will be ?

(A) 2.5 atm (B) 3.7 atm

(C) 1.85 atm (D) 1.3 atm

Q30. The relationship between the values of osmotic pressure of solutions obtained by dissolving 6.00 gL^{-1} of CH_3COOH (π_1) and 7.45 gL^{-1} of KCl (π_2) is ?

(A) $\pi_1 > \pi_2$ (B) $\pi_1 < \pi_2$

(C) $\pi_1 = \pi_2$ (D) None of these

Q31. To 500 cm^3 of water 3.0×10^{-3} kg of acetic acid is added. If 23% of acetic acid is dissociated what will be the depression in freezing point ? K_f and density of water are 1.86 K kg^{-1} and 0.997 gm cm^{-3} respectively :-

(A) 0.186 K (B) 0.228 K

(C) 0.372 K (D) 0.556 K

Q32. 0.004 M solution of Na_2SO_4 is isotonic with 0.01 M solution of glucose at same temperature. The degree of dissociation of Na_2SO_4 is (approx) :-

(A) 50% (B) 25%

(C) 85% (D) 75%

Q33. The freezing point of a 0.05 m $BaCl_2$ in water (100% ionisation) is about ($K_f = 1.86$ km^{-1})

(A) –0.279°C (B) –0.558°C

(C) –0.093°C (D) –0.186°C

Q34. The osmotic pressure of 0.010 M solutions of KI and of sucrose ($C_{12}H_{22}O_{11}$) are 0.432 atm and 0.24 atm respectively. The vanlt Hoff factor for KI is?

(A) 0.80 (B) 1.2

(C) 1.80 (D) 1.0

Q35. If the various terms in the given below expressions have usual meanings, the van't Hoff factor (i) cannot be calculated by which one of the expressions :-

(A) $\pi V = \sqrt{i}\, nRT$

(B) $\Delta T_f = ik_f.m$

(C) $\Delta T_b = ik_b.m$

(D) $\frac{P^0_{solvent} - P_{solution}}{P^0_{solvent}} = i\left(\frac{n}{N+n}\right)$

Q36. 17.4% (W/V) of potassium sulphate (mol. wt. = 174) is isotonic with 4% (W/V) aqueous solution of NaOH. If NaOH is 100% ionised. The degree of ionisation of potassium sulphate is?

(A) 50% (B) 75%

(C) 40% (D) 60%

Q37. Van't Hoff factor for a dilute aqueous solution of HCN is 1.00002. The percent degree of dissociation of the acid is ?

(A) 2×10^{-5} (B) 2×10^{-4}

(C) 2×10^{-3} (D) 2×10^{-2}

Q38. Phenol associates in benzene to a certain extent for a dimer. A solution containing 20×10^{-3} kg of phenol in 1 kg of benzene has its freezing point decreased by 0.69 kelvin. Calculate precentage degree of association of phenol.

(K_f for benzene = 5.12 K.Kg.mol^{-1}) :-

(A) 60% (B) 85%

(C) 73.3% (D) 90.3%

Q39. The Van't Hoff factor 0.1 M La $(NO_3)_3$ solution is found to be 2.74 the percentage dissociation of the salt is ?

(A) 85 % (B) 58 %

(C) 65.8% (D) 56.8%

Q40. If benzoic acid (m.wt= 122) is associated into double molecules when dissolved in benzene and the osmotic pressure of solution of 5gms of benzoic acid in 100 ml benzene is 5.73 atm at 10^0C. What is the percentage association of benzoic acid.

(A) 8 (B) 80

(C) 40 (D) 20

Q41. 0.1 formal solution of NaCl is found to be isotonic with 1.1 % solution of Urea. Calculate the apparent degree of ionisation of NaCl.

(A) 0.083 (B) 8.3

(C) 0.83 (D) 0.42

Q42. Certain substance A tetramerises in water to the extent of 80%. Solution of 2.5 g of A in 100 g of water lowers the freezing point by 0.3°C. The molar mass of A is (K_f = 1.86) ?

(A) 122 (B) 31

(C) 244 (D) 62

Q43. 0.1 molal aqueous solution of an electrolyte AB_3 is 90% ionised. The boiling point of the solution at 1 atm is

(given B.pt. of water = 373 K)

(K_b for H_2O = 0.52K Kg mol^{-1})

(A) 273.19 K (B) 374.92 K

(C) 376.4 K (D) 373.20 K

Exercise - 2A

Brain Booster

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Two liquids A and B form an ideal solution at temperature T. When the total vapour pressure above the solution is 600 torr, the mole fraction of A in the vapour phase is 0.35 and in the liquid phase is 0.7. Then –

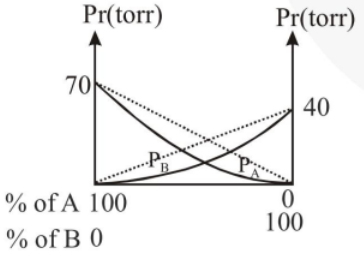
(A) Vapour pressure of pure A at T = 300 torr.

(B) Vapour pressure of pure A at T = 1300 torr.

(C) Vapour pressure of pure B at T = 1300 torr.

(D) Vapour pressure of pure B at T = 300 torr.

Q2. A solution of A(l) & B(l) has following pressure-composition curve, azeotropic composition of A & B has equal mol% of both liquid then select the correct statement(s) about fractional distillation at constant pressure. Molar mass of A & B is 20 & 40 gm/mole.



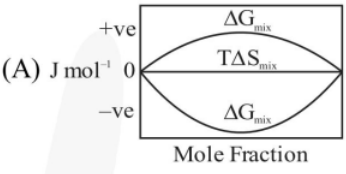
(A) If equal mass % of A & B are distilled then pure A may be obtained in residual liquid.

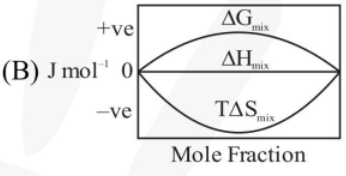
(B) If equal mass % of A & B are distilled then pure A may be obtained in distillate.

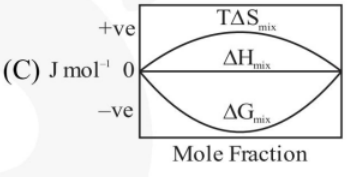
(C) If equal mass % of A & B are distilled then pure B may be obtained in vapours.

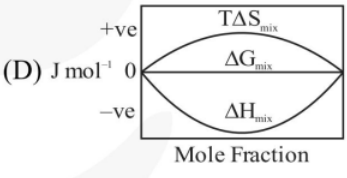
(D) Neither A nor B will be obtained in pure form in residual liquid irrespective of their composition of initial liquid.

Q3. Which of the following represents correctly the changes in thermodynamic properties during the formation of 1 mole of and ideal binary solution ?

(A) 

(B) 

(C) 

(D) 

Q4. Two elements A and B form compounds having molecular formula AB_2 and AB_4. When dissolved in 20g of C_6H_6. 1g of AB_2 lowers the freezing point by 2.3 K, whereas 1.0g of AB_4 lowers it by 1.3 K. The molar depression constant for benzene is 5.1 K kg mol^{-1}. Then -

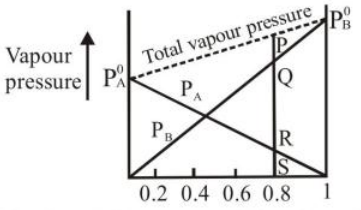
(A) Atomic mass of A = 25.58 u

(B) Atomic mass of B = 42.64 u

(C) Atomic mass of A = 42.64 u

(D) Atomic mass of B = 25.58 u

Q5. Consider the following vapour pressure composition graph, SP is equal to :-



(A) PQ + RS (B) PQ + QR + RS

(C) SR + SQ (D) PQ + QR

Q6. Select correct statements -

(A) Gases which have high value of van der Waals constant 'a' are easily liquefied.

(B) Easily liquefied gases are water soluble.

(C) Ions forming gases in a solvent are soluble in that solvent.

(D) When CO_2 gas is dissolved in water, pressure is decreased.

Q7. Acetone and carbon disulphide form binary liquid solution showing positive deviation from Raoult law. The normal boiling point (T_b) of pure acetone is less than that of pure CS_2. Pick out the **incorrect** statements among the following :

(A) Boiling temperature of mixture is always less than boiling temperature of acetone.

(B) Boiling temperature of Azeotropic mixture is always less than boiling temperature of pure CS_2.

(C) When a small amount of CS_2 (less volatile component) is added to excess of acetone boiling point of resulting mixture increases.

(D) A mixture of CS_2 and CH_3COCH_3 can be completely separated by simple fractional distillation.

Q8. Which of the following forms ideal solution ?

(A) $C_6H_5Cl - C_6H_5Br$

(B) $C_6H_6 - C_6H_5CH_3$

(C) Hexane-heptane

(D) $CHCl_3$-acetone

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for (Q. 9 to 11)

Properties such as boiling point, freezing point and vapour pressure of a pure solvent change when solute molecules are added to get homogeneous solution. These are called colligative properties. Applications of colligative properties are very useful in day-today life. One of its examples is the use of ethylene glycol and water mixture as anti-freezing liquid in the radiator of automobiles.

A solution M is prepared by mixing ethanol and water. The mole fraction of ethanol in the mixture is 0.9.

Given: Freezing point depression constant of water $(K_f^{water}) = 1.86\ K\ kg\ mol^{-1}$

Freezing point depression constant of ethanol $(K_f^{ethanol}) = 2.0\ K\ kg\ mol^{-1}$

Boiling point elevation constant of water $(K_b^{water}) = 0.52\ K\ kg\ mol^{-1}$

Boiling point elevation constant of ethanol $(K_b^{ethanol}) = 1.2\ K\ kg\ mol^{-1}$

Standard freezing point of water = 273 K, Standard freezing point of ethanol = 155.7 K, Standard boiling point of water = 373 K.

Standard boiling point of ethanol = 351.5 K, Vapour pressure of pure water = 32.8 mm Hg, Vapour pressure of pure ethanol = 40 mm Hg, Molecular weight of water = 18 g mol^{-1}, Molecular weight of ethanol = 46 g mol^{-1}

In answering the following questions, consider the solutions to be ideal dilute solutions and solutes to be non-volatile and nondissociative.

Q9. The freezing point of the solution M is -

(A) 268.7 K (B) 268.5 K

(C) 234.2 K (D) 150.9 K

Q10. The vapour pressure of the solution M is -

(A) 39.3 mm Hg (B) 36.0 mm Hg

(C) 29.5 mm Hg (D) 28.8 mm Hg

Q11. Water is added to the solution M such that the fraction of water in the solution becomes 0.9. The boiling point of this solution is -

(A) 380.4 K (B) 376.2 K

(C) 375.5 K (D) 354.7 K

Paragraph for (Q. 12 to 15)

At the critical point, the density of liquid CO_2 is equal to the density of gaseous CO_2. This occurs at specific conditions of temperature and pressure. At temperatures and pressures above the critical point values, CO_2 is deemed supercritical. For a supercritical fluid, the density and ability to dissolve other substances are similar to values expected for liquids.The following figures give phase data for CO_2.

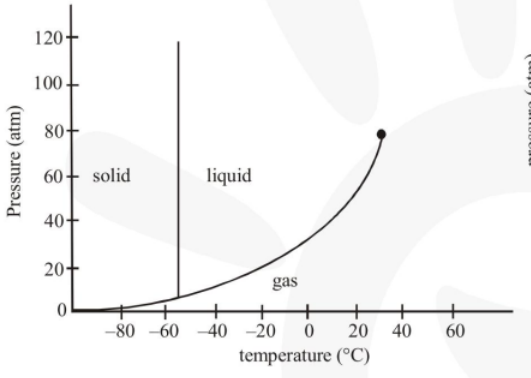


Figure 1 : Phase diagram for CO_2

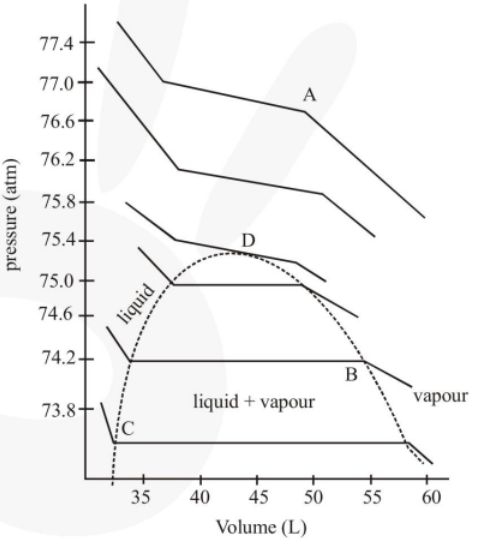


Figure 2 : Isothermal curves for CO_2 near the critical point.

Q12. In Figure 2, which of the points (A-D) is the critical point for CO_2 ?

(A) A (B) B (C) C (D) D

Q13. In an extraction of an organic oil, which of the following is an advantage of using supercritical CO_2 ?

(A) It reacts with most organic compounds

(B) It is easily handled at room temperature

(C) It crystallizes easily.

(D) It is easily removed by evaporation

Q14. Water is a liquid at room temperature, yet CO_2 at room temperature is liquid only at high pressures. Which of the following best explains this?

(A) CO_2 is polar and has strong intermolecular forces.

(B) CO_2 is nonpolar and has strong intermolecular forces.

(C) CO_2 is polar and has weak intermolecular forces.

(D) CO_2 is nonpolar and has weak intermolecular forces.

Q15. According to Figure 1, what is the critical temperature and pressure of CO_2?

(A) – 56.1°C and 6.0 atm (B) – 54.1°C and 119.0 atm

(C) 31.1°C and 75.3 atm (D) 25.0°C and 1.0 atm

Q16. Which of the following compounds is most soluble in supercritical CO_2 ?

(A) NaCl (B) $C_2H_5OC_2H_5$ (C) NH_4NO_3 (D) KOH

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17.

Column-I	**Column-II**
(A) Acetone + $CHCl_3$	(p) $\Delta S > 0$
(B) Ethanol + Water	(q) $\Delta H > 0$
(C) $C_2H_5Br + C_2H_5I$	(r) $\Delta H < 0$
(D) Acetone + Benzene	(s) Maximum boiling azeotropes
	(t) Minimum boiling azeotropes

Q18. Match the column –

Column I (Properties)	**Column II (Affecting factors)**
(A) Relative lowering of vapour pressure	(p) Directly proportional to van't Hoff factor, i.
(B) Elevation in boiling point	(q) Directly proportional to molality
(C) Freezing point	(r) Directly proportional to molarity
(D) Osmotic pressure	(s) Indirectly proportional to lowering of vapour pressure.

Q19. Match the column :

Column I	**Column II**
(A) 0.1 M $Al_2(SO_4)_3$	(p) Solution with highest boiling point
(B) 0.1 M $AlPO_4$	(q) Van't Hoff factor is greater than 1
(C) 0.1 M urea	(r) Solution with lowest osmotic pressure
(D) 0.1 M $MgCl_2$	(s) Solution with lowest freezing point

Q20. Match the column :

Column I	Column II
(A) At constant temperature, the mass of the gas dissolved in a given volume of a liquid is proportional to the pressure of the gas.	(p) Azeotropic solution
(B) If A is solute, B is solvent and d is the density of solution, $\frac{1000M}{1000d - MMA}$ is	(q) Negative deviation from Raoult's law
(C) Liquid mixture which distills over without change in composition	(r) Positive deviation from Raoult's law
(D) CCl_4 and $CHCl_3$ system	(s) Molality
(E) Acetone and $CHCl_3$ system	(t) Henry's law
	(u) van't Hoff law

Q21. Match the column :

Column I	Column II
(A) K_f	(p) $P°x_2$
(B) K_b	(q) $\frac{RT}{V°}\ln\left(\frac{P°}{P}\right)$
(C) $P° - P$	(r) Elevation of boiling point in one molal solution of nonelectrolyte
(D) p	(s) $\frac{RT_f^2}{1000\ell_f}$

Assertion and Reason

The following questions 16 to 20 consists of two statements each, printed as Assertion and Reason. While answering these questions you are to choose any one of the following five responses.

(A) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

(B) If both Assertion and Reason are true but Reason is not correct explanation of the Assertion.

(C) If Assertion is true but the Reason is false.

(D) If Assertion is false but Reason is true.

(E) If Assertion & Reason are false.

Q22. **Statement 1 :** If a liquid solute more volatile than the solvent is added to the solvent, then the vapour pressure may increase i.e. $P_s > P°$.

Statement 2 : In the presence of a more volatile liquid solute, only solute will form the vapours and solvent will not.

Q23. **Statement-I :** The freezing of water is an endothermic process.

Statement-II : Heat must be removed from the water to make it freeze.

Q24. **Statement-I :** Addition of ethylene glycol (non-volatile) to water lowers the freezing point of water hence used as antifreeze.

Statement-II : Addition of any substance to water lowers its freezings point.

SUBJECTIVE TYPE

Q25. Find out the molarity of 1 litre of 93% (w/W) H_2SO_4 (density = 1.84 g/ml).

Q26. The vapour pressure of benzene at 90^0C is 1020 torr. A solution of 5 g of a solute in 58.5 g benzene has vapour pressure 990 torr. The molecular weight of the solute is?

Q27. A 5% solution of anhydrous $CaCl_2$ at 0^0C developed 15 atm. Osmotic pressure. What is the degree of dissociation of $CaCl_2$.

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
B	A	C	A	D	A	B	A	A	C
11	12	13	14	15	16	17	18	19	20
A	A	C	C	C	C	C	B	A	D
21	22	23	24	25	26	27	28	29	30
D	C	D	D	C	D	C	B	B	C
31	32	33	34	35	36	37	38	39	40
B	C	C	B	B	B	D	C	C	C
41	42	43	44	45	46	47	48	49	50
B	C	B	C	B	D	D	B	A	A
51	52	53							
C	A	A							

Ex-1A

1. A, B, C, D **2.** A, C **3.** A, D **4.** B **5.** B

6. B **7.** B **8.** A **9.** B

10. A-P ; B-P ; C-Q,S ; D-Q,R,S **11.** (A) → q ; (B) → p ; (C) → r, s ; (D) → p, t

12. A-Q ; B-P,S,T ; C-R,U ; D-R **13.** A → p q, B → p r ,C→ p s, D → P

14. A-s ; B-r ; C-q ; D-p **15.** M = N = 0.25

16. 4.5 N **17.** 3.65 gm/lit

18. (i) P_{CH_3OH} = 60 mmHg (ii) Y_{CH_3OH} = 0.78 mmHg

$P_{C_2H_5OH}$ = 17 mmHg $Y_{C_2H_5OH}$ = 0.22 mmHg

P_S = 77mmHg

19. 0.8 **20.** X_B = 0.74 **21.** P_S = 43.2 mmHg

22. $T_B^o(Br_2)$=332.195K **23.** K_b = 1 **24.** M_{wt}=105.68

25. M_{wt} = 34.89 **26.** C_2 = 0.061 M

Ex-2

1	2	3	4	5	6	7	8	9	10
A	C	B	B	B	C	A	A	B	C
11	12	13	14	15	16	17	18	19	20
A	C	D	D	A	A	B	C	C	B
21	22	23	24	25	26	27	28	29	30
A	D	D	A	C	D	B	D	C	B
31	32	33	34	35	36	37	38	39	40
B	D	A	C	A	A	C	C	B	B
41	42	43							
C	D	D							

Ex-2A

1. A, C	**2.** B, D	**3.** C	**4.** A, B	**5.** B, C
6. A, B, C	**7.** A, C, D	**8.** A, B, C	**9.** D	**10.** B
11. B	**12.** D	**13.** D	**14.** D	**15.** C

16. B

17. (A) → p, s, r ; (B) → p, q, t ; (C) → p ; (D) → p, q, t

18. (A → p; B → p, q; C → s; D → p, r)

19. (A → p, q, s; B → q; C → r ; D → q)

20. (A → t; B → s, C → p, D → r, E → q)

21. (A → s; B → r, C → p, D → q)

22. C	**23.** D	**24.** C	**25.** 17.5 M	**26.** 220

27. 24.5%